

Optimal Planning of Battery Swapping and Charging Stations for the Urban Cities

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Abstract— Electric mobility is becoming necessary for urban zones with ever-increasing fuel costs and rising air pollution levels. Location planning of Battery Swapping and Charging Station (BSCS) for battery electrical buses (BEBs) requires to consider many factors. The existing research focuses on social cost model, vehicle relocation problem and local charging stations. However, battery swapping stations with charging stations, and urban routes have not yet been studied for BSCS to charge BEBs. In order to solve above problem this paper presents a selection of sites for optimal placement of battery swapping and charging stations in urban cities considering the routes and location of bus depots for battery electric buses. The optimal location of BSCS for BEBs has the effect of reducing the overall investment cost and charging cost. Secondly, this paper establishes a BSCS location optimization model which simplifies into a road network to meet the maximum accessibility of BSCS to the BEBs. Finally, to identify the optimal location for BSCS, minimum total construction cost of BSCS considering land price and charging station is discussed in this article. The real-world road networks of Pune city are taken for parameter analysis.

Keywords— *Electrical Vehicle, Battery Swapping and Charging Station, Battery Electric Bus, Location Optimization.*

I. INTRODUCTION

Now a days the ownership of electric vehicles (EV) has been increased rapidly due the support of national policies and increasing fuel prices. The mid-twentieth century, latent increase in pollution of environment has grown on the planet in an accelerated manner. Presently, the major source for conventional vehicles is fossil fuels like diesel or gasoline. In the future fossil fuels will run out still the consumption of petroleum fuel is increasing rather it should decrease. Increasing air pollution due to increasing demand of conventional vehicles leads to an increase of the concentration of carcinogenic substances in the air such as Nitrogen Dioxide (NO₂), Carbon Dioxide (CO₂) and Sulphur Dioxide (SO₂) by extensive quantity. Indian capital Delhi is one of the 21st cities among worlds most polluted 30 cities in the year 2019. The diesel buses are still dominant in the public transport, and which produce 75% of matter emission and 45% of Nitrogen Dioxide [1]. To solve this problem, the world needs to take strong action and others needs to adopt more EV. In many countries, governments have launched policies as taken primary objectives to reduce said environmental problem. along with this, the government have proposed various strategies to integrate electric vehicles into the transportation system. Battery electric buses introduced by various governments as part of transportation electrification plan. The

advancement of battery and rapid increasing technologies in bus manufacturing process, a significant attention is given world widely. Some initiatives have been taken to promote green transportation, like in China ten cities and thousand vehicles program, in the United States TIGER Program, in German the Electricity Mobility program and in UK Green bus fund program [2]. The motivation of these government will lead extensive use of BEBs in metropolitan cities. Many cities in the US introduced BEBs in mid 2000s. Uruguay has signed the deals of BEBs in 2012.

BEBs usually have fixed depots, fixed running, and approximately the same battery capacity. Swapping policy decisions or potential battery recharging, charging station siting, sizing decisions, and battery sizing are the challenges while configuring public transportation systems for BEBs [3]. BEBs still suffers from insufficient charging problem, limited mileage ranges from long charging time as compared with conventional vehicles regardless of its regenerative energy during the braking process. BEBs the range of BEBs significantly reduced due to driving behavior, air conditioning and battery degradation issues therefore with single charge BEBs not able to do all days' work [4].

Nowadays, fast charging, slow charging, and battery swap are the three charging methods used by many users to charge their EVs. BEBs utilization is minimized due to the slow charging station, and it takes hours to refuel it. Fast charging can reduce charging time by $\frac{1}{4}$ but it impacts on the battery life [5][6]. EV chargers cost from \$1,000 to \$1,00,000 and it depends on the type of charger. Many efforts have taken by the researcher to deploy battery electric buses [7] [8]. It has been discussed that EV adoption is a necessity for realizing its environmental gains with well-aligned charging strategies. Otherwise, generation costs will be increased as erratic charging by the users. EV users are required to charge them by considering the power grid [9].

Increasing the profit of BSS is very needful, therefore BSS stations need to know when to charge batteries and which is the best location for BSS. Charging of the batteries, a novel centralized optimal model is proposed based on the charging strategy of BEBs including electricity price, and also using a radial distribution system an optimal charging station location determined [10]. Optimal charging station locations along with the optimal charging priority of BEBs are developed based on a heuristic algorithm [11]. In 2013, Tesla Motors Company demonstrated the battery-swapping technology. This battery swap technology was developed by Tesla that can

take only nearly 80 to 90 seconds to exchange discharged batteries with a charged one [12]. Meanwhile, different techniques were developed to swap the battery for public transport [13].

The price of electricity is based on historical data from swapping stations along with the electrical demand [14]. the infrastructure planning process for the BSS location was developed with a robust model with planning strategies and the impact of batteries [15] but, there is to consider the public transport bus scenario and routes of the city. Furthermore, operating charging station models were proposed for the study of electrical price along with the demand for batteries in [16]. To assign proper charging scheduling of BSS and to reduce the operational cost of electricity price optimal model was developed by Wu [17].

Many researchers have considered battery swapping stations only or else they have considered the charging stations only to solve problems related to the stations. As in the market, both types of vehicles are coming so there is a need to consider both simultaneously. It will take a few more years to integrate both types of chargers. May governments are planning to use BEBs, or they are moving towards it in their countries. So now researchers have a broader scope to solve problems related to the same. The sizing of the BSS and the location for the BSS were solved by formulating a mixed-integer nonlinear and non-convex programming [18]. Many authors proposed the optimal placement of different systems, not only BSS but also like PMUs [19].

A combined road and grid networks considering robust optimization model was proposed to solve location problem of electrical vehicle charging stations [20]. Battery swapping and charging stations are needed to consider to utilize the BEBs by finding the optimal locations. The optimal placement of various systems based on the charging scheduling for reducing the price by matching the demand has been solved by proposing a generalized Benders decomposition-based algorithm in [21]. Focusing on urban Electric taxis, an optimized battery-swapping station scheme (BSS) is proposed [22]. The location planning of charging stations and battery-swapping stations takes into account the behavioral capabilities of the users [23]. Considering Random BEB arrivals and chaotic charging behavior, a two-level model was proposed to estimate the energy utilization [24]. A two-stage stochastic model is presented to handle demand uncertainties of BEBs for BSS [25]. Again, there is a need to consider the battery swapping station for BEBs along with the charging station for investment cost maximization.

The nature of charging station's location choices are deliberate. Locations of bus depots or charging station for BEBs remains same for many years. The future operations of the BSCS are not yet clear, along with what policies and conditions will be defined. Battery swapping operation may vary according to the condition of weather and the transportation network. The total cost of the battery and charging station needs to be lower with technological advancement. The Government companies are the major investors for building infrastructure for the BEBs as the urban buses are going to be electrified. The government is going to decide the location of BSCS and configure the different factors associated with it. The aim of this paper is to simplify

the location model for placement of battery swapping and charging station together.

However, no study has investigated the optimal location based on the city routes, investment of charging piles and swapping stations for the battery electric buses. Other constraints may affect the optimal location for BSCS but to a lesser extent, so the main constraints considered are investment and charging cost. The investment and charging cost are proposed considering the application to the Government Battery Electric Buses. This paper presents an optimal location for BSCS considering the route of BEBs and optimal number of charging station placement. The paper has been organized as follows. In section II, mathematical problem formulation is presented. In section III, how model can solve the given problem is presented. In section IV, the system model is constructed considering route locations and case study. In section V, the optimal placement of battery swapping and charging stations results are presented.

II. PROBLE FORMULATION

In order to identify optimal planning and building BSCS, we have proposed a total cost model. The cost model is mainly based on the BSCS investment cost and charging cost. The investment cost includes the land price for building BSCS, charging station pile cost, and the swapping station cost. Charging cost includes the electricity price over 24 hours and demand.

The objective function can be represented by equation (1)

$$\min F = C_{inv} + C_{charge} \quad (1)$$

where F represents the total cost of the system. C_{inv} indicates the investment cost of the system. C_{charge} gives the charging cost at the BSCS. Both depend on the scale of the charging stations, reflected by the number of charging stations to be installed. The construction cost of one charging station is for one location including the swapping station cost, charging pile cost and land cost. i.e., construction cost of BSCS is represented as equation (2).

Thus, the following formula can be obtained.

$$C_{inv} = \left[\frac{d \cdot (1 + d)^{Yp}}{d \cdot (1 + d)^{Yp} - 1} \right] \sum_{i \in I} X_i Ld_p + Nc_i Cp + Ns_i Cs \quad (2)$$

The charging cost consists of charging of Electrical Bus and charging of swapped batteries over the period of time as shown in the following equation (3).

$$C_{charge} = 365 \left[\sum_{i,t \in Itp} d_i^t B_{i,t} S_{i,t} \right] \quad (3)$$

Constraints:

$$X_i + \sum_{i \in I, j} X_j \leq \beta_i \quad \forall i \in I \quad (4)$$

$$\beta_i = 1 \quad (5)$$

$$\sum_{i \in I} X_i \leq Ncs \quad \forall i \in I \quad (6)$$

$$d_i = \sum_t d_i^t \quad (7)$$

$$x_i \in \{0,1\} \quad (8)$$

Constraint (4) and (5) represent where to place charging station in the given traffic network. The binary decision variable Beta i is taken to select the location. The given equations must choose minimum locations from the set of location I. Equation (6) is responsible for the number of charging stations to build at selected location. Equation (7) indicated the charging demands at selected sites. Also, it indicates the number of vehicles that can be charged in the charging station over a period of 24 hours. The charging demand is indicated by the d_i^t at the demand areas for each interval of time period t. Equation (11) is the binary decision variable which is set i if charging station is placed or 0 if charging station is should not be placed at location i.

The total cost model is developed with above all equations. This constructed cost model effectively measures investment and charging cost. This model calculates the total cost and considering different scenarios as mentioned above. The objective function is solved using GAMS software.

Along with the constraints some assumptions are made before formulating the problem.

i) all the battery electrical bus are owned by the Maharashtra State Road Transportation Corporation (MSRTC); II) all the batteries are owned by the installed station; III) charging facilities are available at BSCS; iv) all buses travel in the given route; v) all battery used in BEB are uniform type. Also, EVs can swap batteries or charge the battery to the nearest station.

Parameter description: X_i - either to build charging station at the location of I; L_{dpu} - the land price in rs; N_{ci} - quantity of the charging pile to built on the network; N_{si} - quantity of the swapping stations to build on the network; d - discount rate; Y_p - year of planning; C_p - price of one charging pile; C_{ss} - price of one swapping station; w - power consumption factor; β_i - binary parameter; d_{ti} - charging demand at point i over the time t; $B_{i,t}$ - number of BEBs and batteries charges at i to t; $S_{i,t}$ - the shortest distance from i to t. The set i consists of locations at different places in Pune city.

III. SOLUTION METHODOLOGY

In order to solve the mathematical model described in section II, a proposed method can be used. The set of all the locations is needed to identify the best optimal location for the BSCS. The proposed methodology for selection of optimal location for BSCS and charging and investment cost as shown in fig. 1. Initially all the sets were defined, for all location i and for time t. The price of charging station pile C_p , swapping station N_s and land price taken. Optimal location model can identify the minimum location where BSCS can be installed for BEBs. The price for each hour is taken from the Indian Energy Exchange.

Initially all the sets were defined for the time and routes. The set I consists of Kothrud, Katraj, Phursugi, Hadapsar, Swargate, Shivaji Nagar, Mundhava, Manjari, Khadki, Viman Nagar, Vishrantwadi, Baner, Bhosari, Lohagaon. The numbering to the locations are started from the Kothrud and ends to Lohagaon. Hence the number one is given to the Kothrud, two is given to Katraj and so on. Then parameters of routes, the number of charging stations at different locations, the number of swapping stations at different locations, the price at different times, and the demand at given time t for the different locations are considered. Some assumptions are made to select number of swapping and charging stations for each location according to the bus depot and quantity of BEBs. Then locations are mapped with each other according to the routes. Then the identified the number of minimum locations where BSCS can be placed. Once the minimum number of charging stations obtained then total investment and charging cost can be calculated for the selected locations. The process flow is shown in fig. 1.

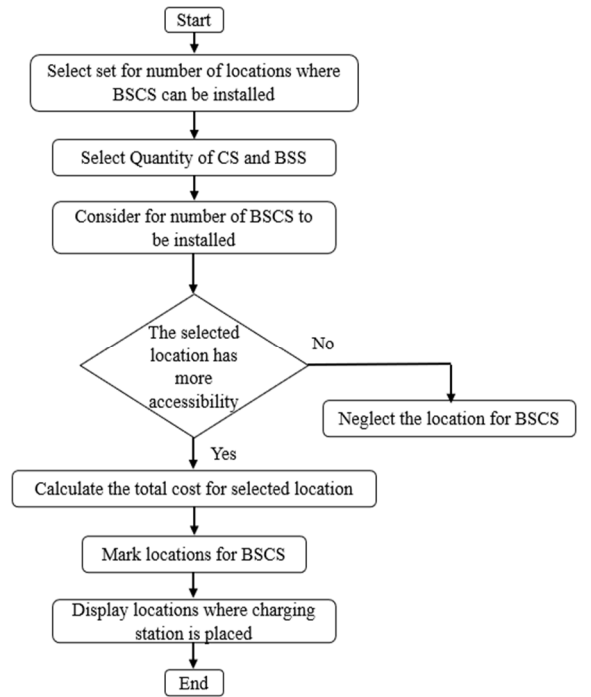


Fig. 1. Process for identifying total cost and location of BSCS.

IV. SYSTEM MODEL CONSTRUCTION

A. Transportation Network Model

The basic structure of location on the simple transportation network is shown in fig. 2. Most visited locations are shown in the circle and vehicle routes are shown by the lines. Mainly the vehicle travels from one location to another location through the road network. The routes are considered where most vehicles travel during the daytime and nighttime. A set of 6 locations are considered for the example. The locations of areas have been shown where a minimum number of BSCS can be placed in traffic network. In the said example the maximum routes are connected to the 5th Location after that 3rd and fourth. 5th area can be accessed by many vehicles and this Location is the best location. Also 3rd and 4th have 3 routes connected here optimization technique can identify the best observable area and where BSCS can be placed.

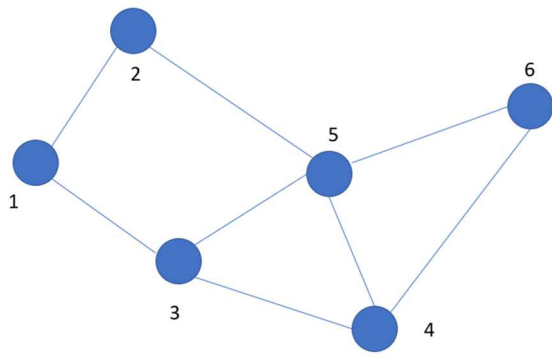


Fig. 2. Road Network Model.

B. Case Study Model

The city of Pune, India ($18^{\circ} 31' 0.2136''$ N, $51^{\circ} 22.5180''$ E) has been taken as a case study for research work. Pune is a city in the western Indian state of Maharashtra, and it has a rich and varied history that dates back over 1600 years. The city was once the seat of the Maratha Empire and played a significant role in the Indian independence movement. Today, Pune is a vibrant and modern city, known for its education and IT industries, as well as its cultural heritage and vibrant street food scene. Also, the Pune city is the first Indian city to adopt e-buses. A total up to 139 e-buses were deployed as of 2019. The map for Pune city with optimal results is shown in fig. 3.

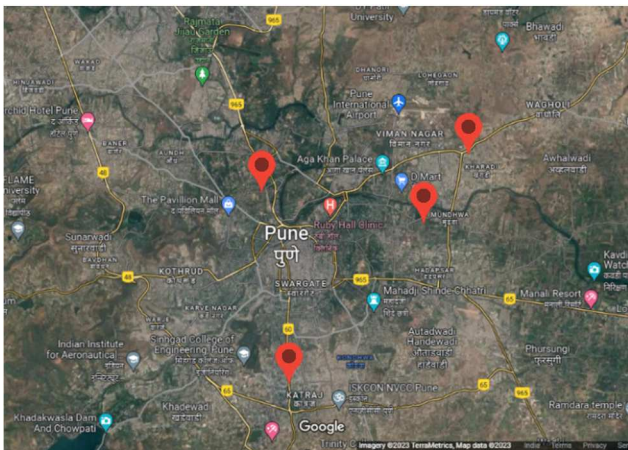


Fig. 3. Pune City Map with Optimal Locations.

In this city 14 nodes are considered as locations where BSCS can be installed with maximum observability and with a minimum number of charging stations. Total 14 locations are considered as set $I = \{1, 2, 3, 4, \dots, 14\}$ in between battery electric buses can travel. The main 14 locations are taken from Pune city and the roads connecting the given locations are shown as straight lines. The simplified route map of Pune city is shown in fig. 4. The locations linked with the given set start from Kothrud and end with the Lohagaon. In the given network the Kothrud is connected to a Baner, Swargate and Katraj. Similarly, all the locations are connected to each other through a road link. It is also assumed that all the time battery can be swapped in depots when the charging of the battery goes to zero. Another assumption is that all BEBs travel in the given region. The actual roads of the city are not a straight line as shown in the fig. 2 and fig. 4. The land cost per unit area may vary as per the location but in this case, the constant cost is considered for calculations.

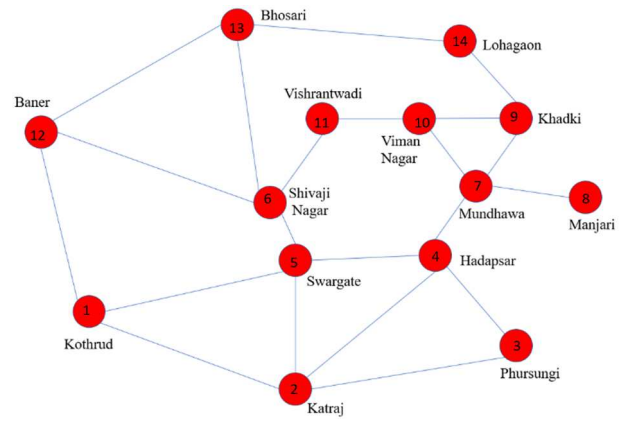


Fig. 4. Pune City Road Network.

V. RESULTS AND DISCUSSION

The CPLEX solver is used to solve current objective functions. Initially one example is taken as the road network model. The road network model was presented in system model construction section IV. The algorithm is applied on the road network model, the 5th and 1st location are obtained as optimal locations for the BSCS where BSCS can be placed. The 5th node is connected to four nearby locations. As there is no road between locations 1 to 5 there are needed two locations for the BSCS. The results are shown in table I.

TABLE I. LOCATIONS FOR BSCS OF EXAMPLE

Sr. No.	Location Name	Number of BSCS Selected
1	1	1
2	2	.
3	3	.
4	4	.
5	5	1
6	6	.

The number of charging stations I_{cs_i} at different locations are considered. In a same way number of swapping stations I_{ss_i} also considered. Here, we have considered Hadapsar, Swargate, Shivaji Nagar, Viman nagar, Mundhawa, and Vishrantwadi are more crowded areas among all the areas. i.e. we have taken extra quantities for the charging station and swapping stations as compared with the other locations as shown in table V. Price for the one charging pile is taken Rs. 5,00,000/-. The price for the one swapping station is considered Rs. 20,00,000/-. And the land price is considered as the Rs. 2,00,000/- for each BSCS. The land price may vary as per the locations to location, but here we have assumed same price for land in Pune city. The considered prices are shown in the table III.

TABLE II. PARAMETERS FOR BSCS

Parameter	Amount in Rs.
C _p - price per charging pile	5,00,000/-
C _s – price per swapping station	20,00,000/-
L _{dp} – Land price per BSCS	2,00,000/-

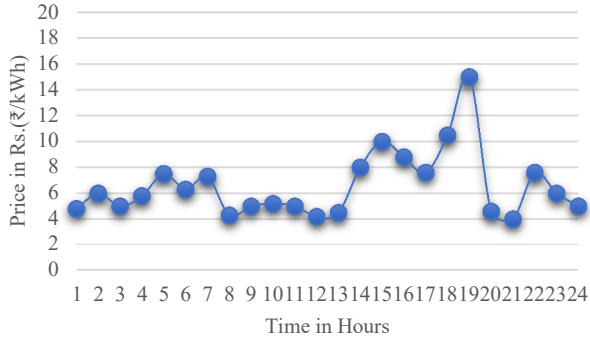


Fig. 5. Electricity price over 24 hours.

The electricity price over 24 hours is taken from the Indian Energy Exchange [26]. In fig. 5, the price is shown with respect to each hour. The price is given in Indian rupees (₹) for the per unit (kWh). This price is used to calculate the charging cost for over one year. The set T is taken for 24 hours' time. Price is considered over in rupees at each time t. Similarly, the demand in kWh is taken for finding charging cost. The random values are taken for the demand. The charging cost for over one year is Rs. 1,59,05,386/-. Along with this we have considered battery/charging BEBs equal to 10 numbers and it is charging during the high demand with low price.

TABLE III. POSSIBLE CHARGING STATIONS

Sr. No.	Location Name	N _c (i)	N _s (i)
1	Kothrud	2	1
2	Katraj	2	1
3	Phursugi	2	1
4	Hadapsar	5	2
5	Swargate	5	2
6	Shivaji Nagar	5	2
7	Mundhava	5	2
8	Manjari	2	1
9	Khadki	2	1
10	Viman Nagar	5	2
11	Vishrantwadi	5	2

Sr. No.	Location Name	N _c (i)	N _s (i)
12	Baner	2	1
13	Bhosari	2	1
14	Lohagaon	2	1

The said algorithm is applied to the Pune city road network. The network of the city is shown in fig. 4. The total 4 locations were found to be optimal locations where BSCS can be installed with maximum observability for the BEBs. These 4 locations are Katraj, Shivaji Nagar, Mundhava and Khadki. Also, the number of locations can be minimized by varying N_cs parameter.

The total investment cost of the BSCS is 5.64 Cr, it includes the land cost as well. The total annual charging cost of battery swapping station is Rs. 1.59 Cr. The cost comparison of the proposed methodology with and without is shown in table IV. The proposed methodology also confirms 100% accessibility for each location.

TABLE IV. RESULTS OF INVESTMENT COST

Method	Selected Locations	Cost in Rs. (₹)
With Algorithm	Katraj, Shivaji Nagar, Mundhava, Khadki	5,64,00,000.00
Without algorithm	Hadapsar, Swargate, Shivaji Nagar, Mundhava, Viman Nagar, Vishrantwadi	8,46,00,000.00

The total cost of BSCS at four different locations with the proposed algorithm is Rs. 7,23,05,386.00/-. The total cost includes the investment cost and charging cost. The investment cost of considered BSCS at four different locations are Rs. 5.64 Cr. Also, the charging cost over one year is Rs. 1.59 Cr.

TABLE V. LOCATIONS FOR BSCS

Sr. No.	Location Name	BSCS placement
1	Kothrud	.
2	Katraj	1
3	Phursugi	.
4	Hadapsar	.
5	Swargate	.
6	Shivaji Nagar	1
7	Mundhava	1
8	Manjari	.

Sr. No.	Location Name	BSCS placement
9	Khadki	1
10	Viman Nagar	.
11	Vishrantwadi	.
12	Baner	.
13	Bhosari	.
14	Lohagaon	.

Initially all sets are defined to calculate the total cost of the charging station. All parameter values are taken into account. Some parameter values are assumed as discussed in above sections, and some are taken from the reference papers. Constraints (4) to (6) are used to identify the optimal location and it gives the exact quantity of charging stations to be installed on the road network. By changing the value of Ncs the number of charging stations locations can be changed in order to balance requirements. Most observable locations are identified to install the battery swapping and charging station. This location can be accessed by nearby BEBs which are traveling through given routes. At the end the total cost of BSCS for BEBs is calculated.

In this paper, the total cost model is presented. The cost model consists of investment cost and the second charging cost. Methodology proposed in this article gives the accurate results as discussed above. This article gives brief information about the planning of battery swapping and charging stations for an urban city. This research article has taken Pune city as an example for application and analysis purpose. This model can be applied to other cities as well.

Future research may include energy strategies for the same BSCS along with considering the renewable, and microgrid integration to the transmission network. Also, in the future, research on the battery degradation factors may consider.

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